

Appendix C2 - Ningaloo Marine Park (National Park Zone) Data Analysis Report

North-west Marine Parks Network

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Data Analysis

Statistical methods and model selection results underpinning the natural values reported in Appendix A2.

Benthic ecosystem extent

Table C2.1: Habitat model selection results. Top generalised additive models (GAMs) for predicting the probability of occurrence of key habitats from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the most parsimonious model (fewest variables - shown in bold) within two units of the lowest AICc. Survey year was offered as a candidate predictor and retained in every top model, so covariate relationships are fitted separately for 2019, 2021, 2022 and 2024.

Response	Model	ΔAICc	ωAICc	R^2	EDF
Sand	detrended+roughness+Z+year	0	1	0.19293	26.77
Sessile invertebrates	detrended+roughness+Z+year	0	1	0.18836	26.04
Reef	detrended+roughness+Z+year	0	1	0.19293	26.77

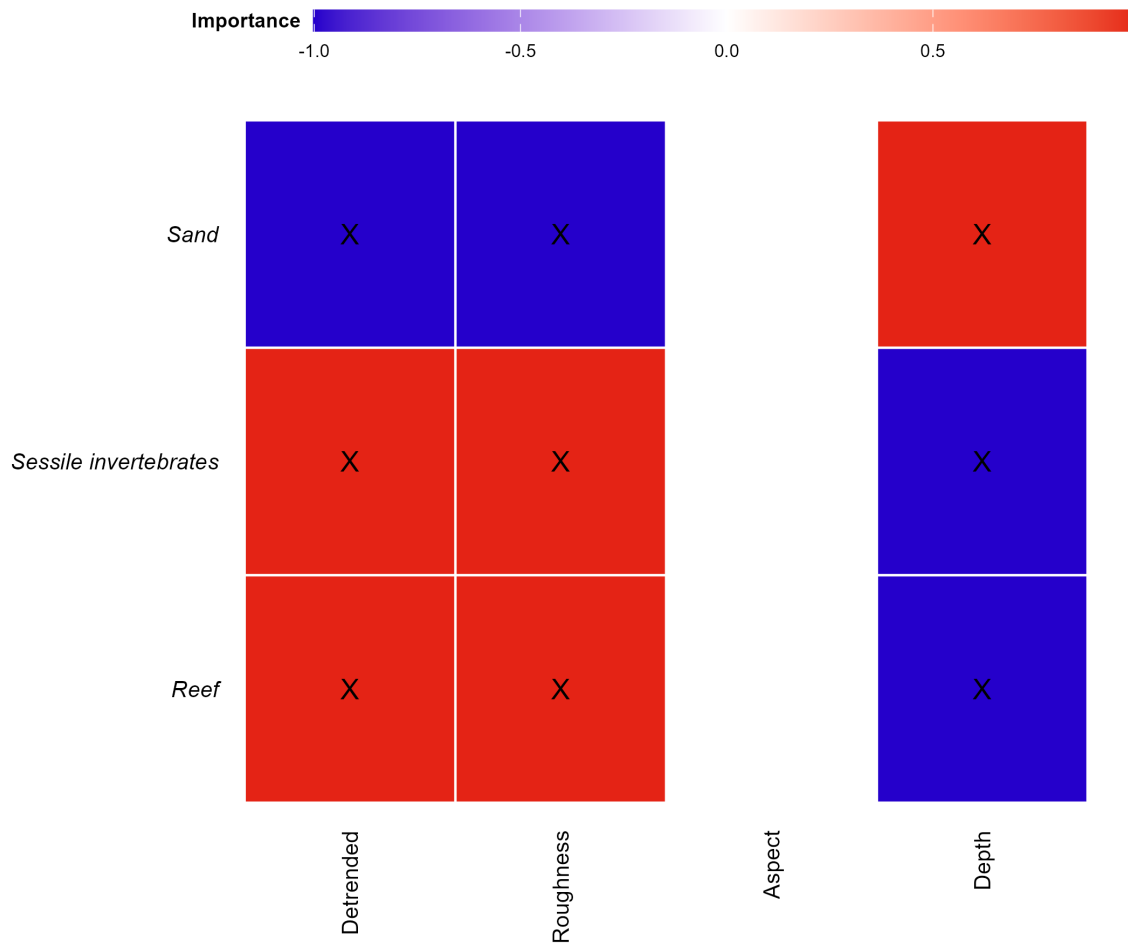


Figure C2.1: Variable importance scores for predicting the probability of occurrence of main habitat categories. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient.

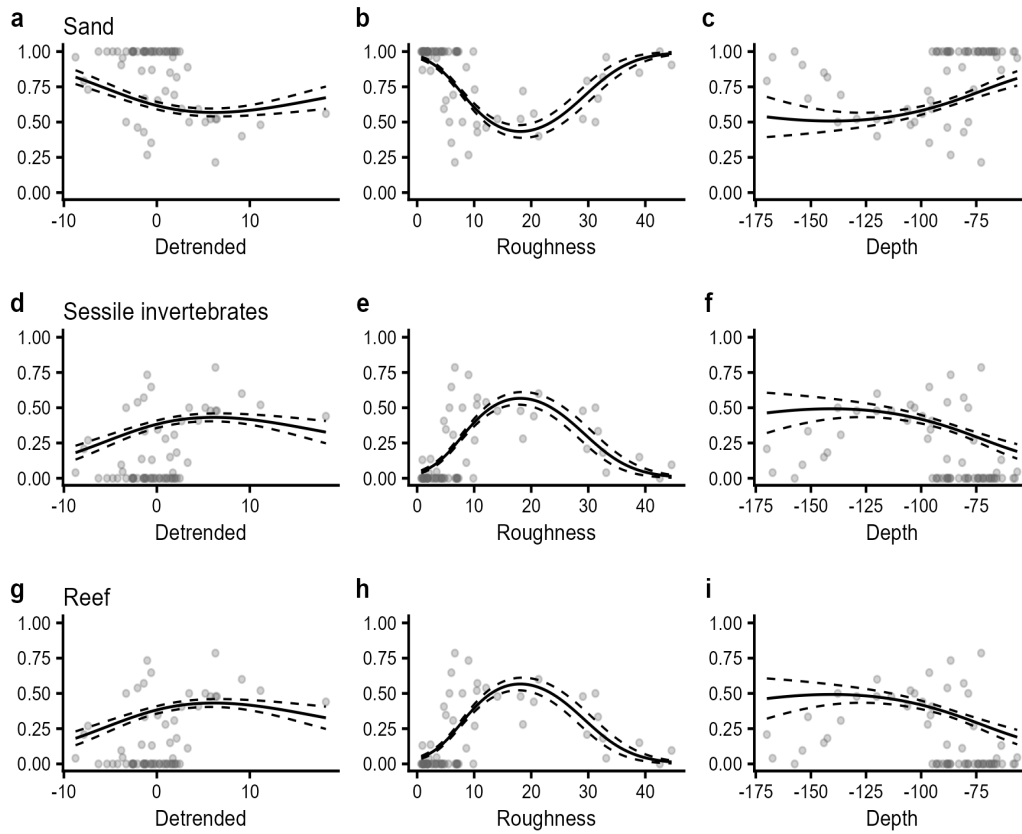


Figure C2.2.1: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types in 2019. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value and survey year held at 2019. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data for that year.

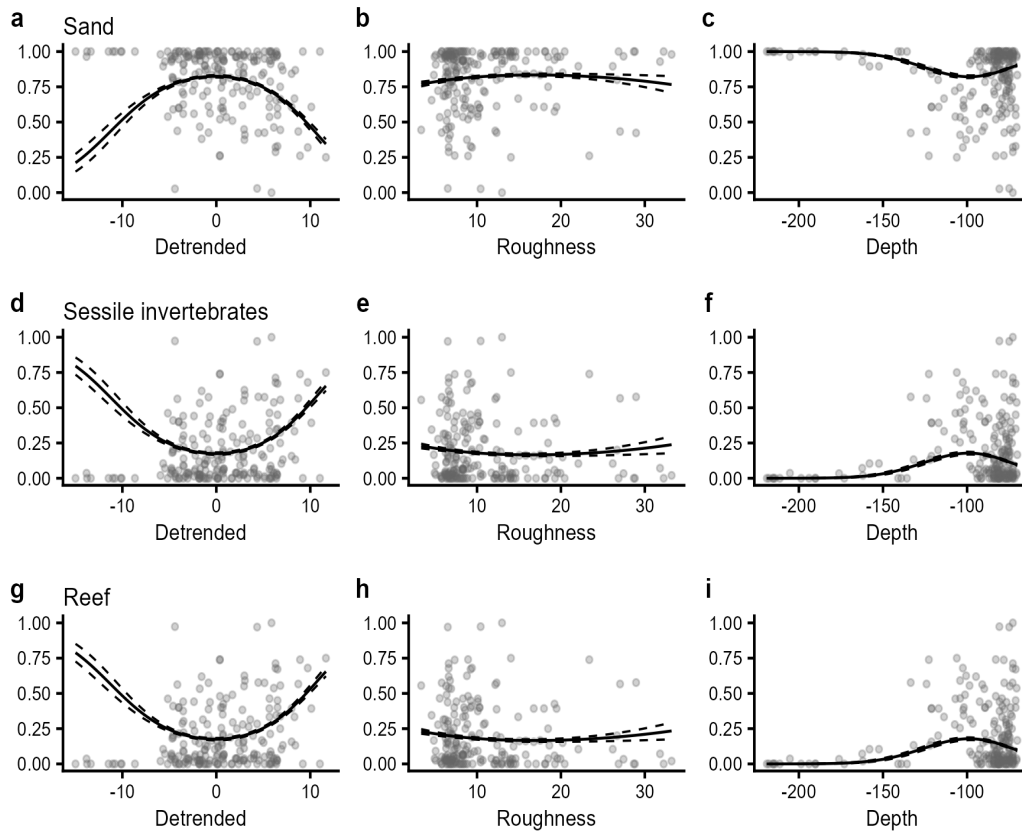


Figure C2.2.2: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types in 2021. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value and survey year held at 2021. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data for that year.

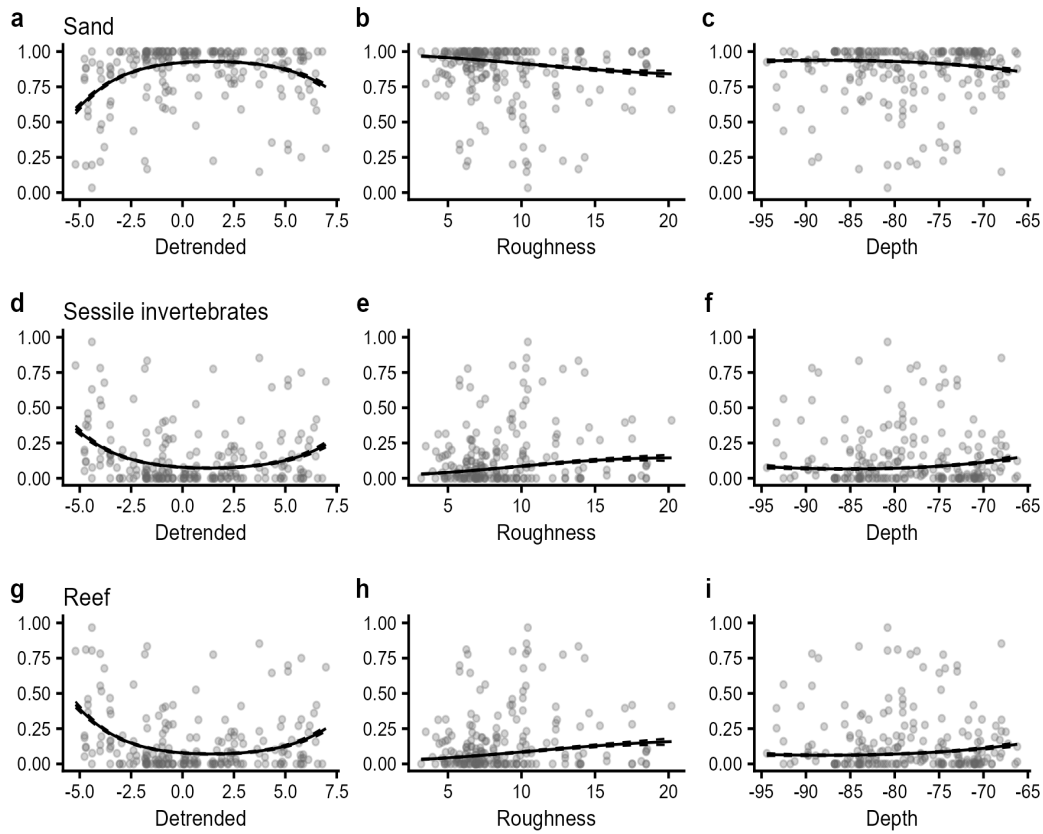


Figure C2.2.3: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types in 2022. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value and survey year held at 2022. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data for that year.

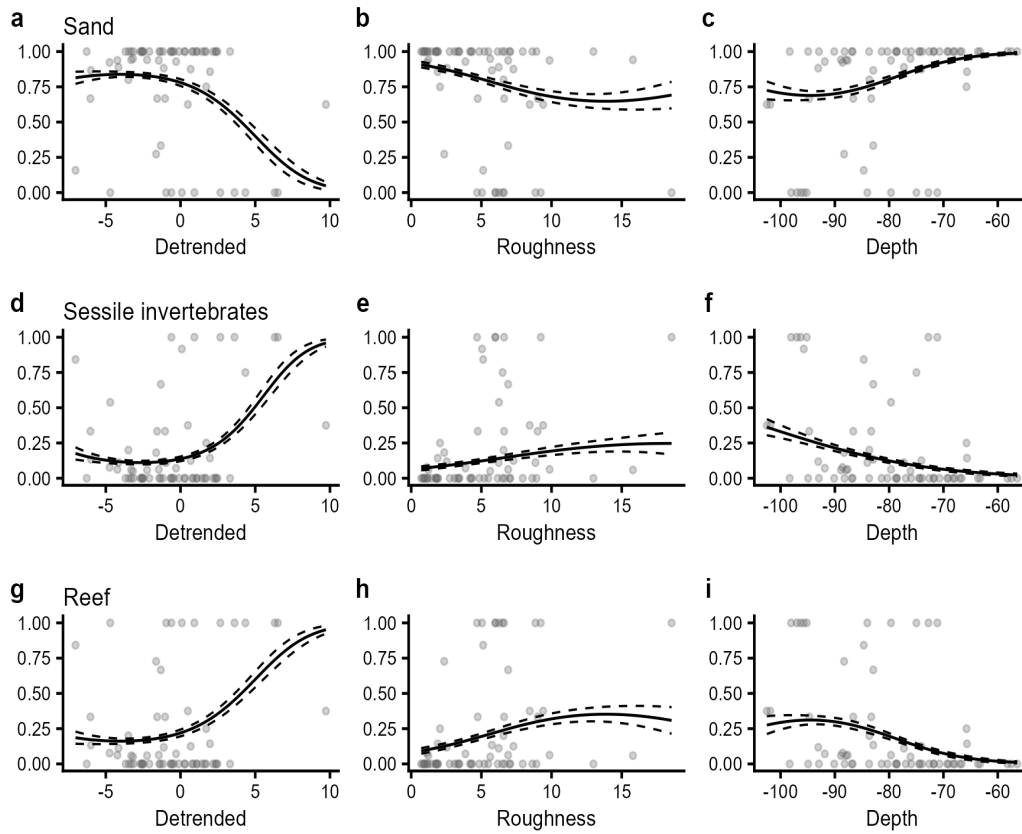


Figure C2.2.4: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types in 2024. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value and survey year held at 2024. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data for that year.

Fish

Table C2.2: Demersal fish model selection results. Top generalised additive models (GAMs) for predicting key fish metrics from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the most parsimonious model (fewest variables - shown in bold) within two units of the lowest AICc. Survey year and zone status (No-Take/Fished) were forced into every candidate model rather than offered as candidates themselves, so the fitted relationships shown for 2019, 2021, 2022 and 2024 share the same covariate effects and differ only by year and status.

Response	Model	ΔAICc	ωAICc	R^2	EDF
Species richness	aspect+Z+reef	0	0.726	0.31291	8.81
	detrended+aspect+Z+reef	1.987	0.269	0.31358	9.82
Total abundance	Z+reef	0	0.336	0.16723	8.13
	aspect+Z+reef	0.579	0.252	0.1749	9.16
	detrended+Z+reef	0.985	0.205	0.17645	9.64
	detrended+aspect+Z+reef	1.249	0.18	0.18473	10.52
Community Thermal Index	aspect+Z	0	0.229	0.28638	7.58
	Z	0.063	0.222	0.2772	6.60
	detrended+aspect+Z	0.725	0.159	0.29015	8.41
	detrended+Z	0.989	0.14	0.28019	7.46
Large Reef Fish Index*	aspect+Z+reef	0	0.54	0.30507	9.36
	detrended+aspect+Z+reef	1.6	0.243	0.30826	10.46

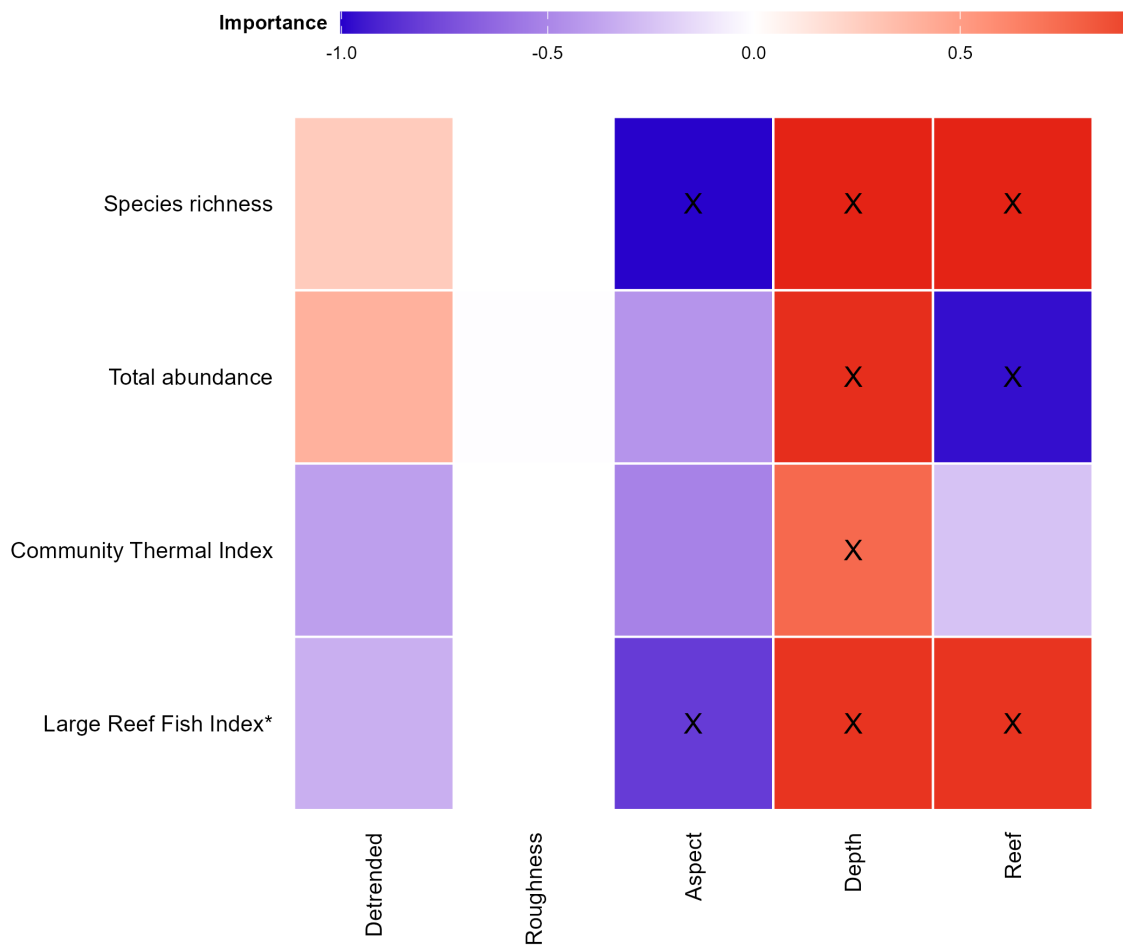


Figure C2.3: Variable importance scores for predicting key demersal fish metrics. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient.

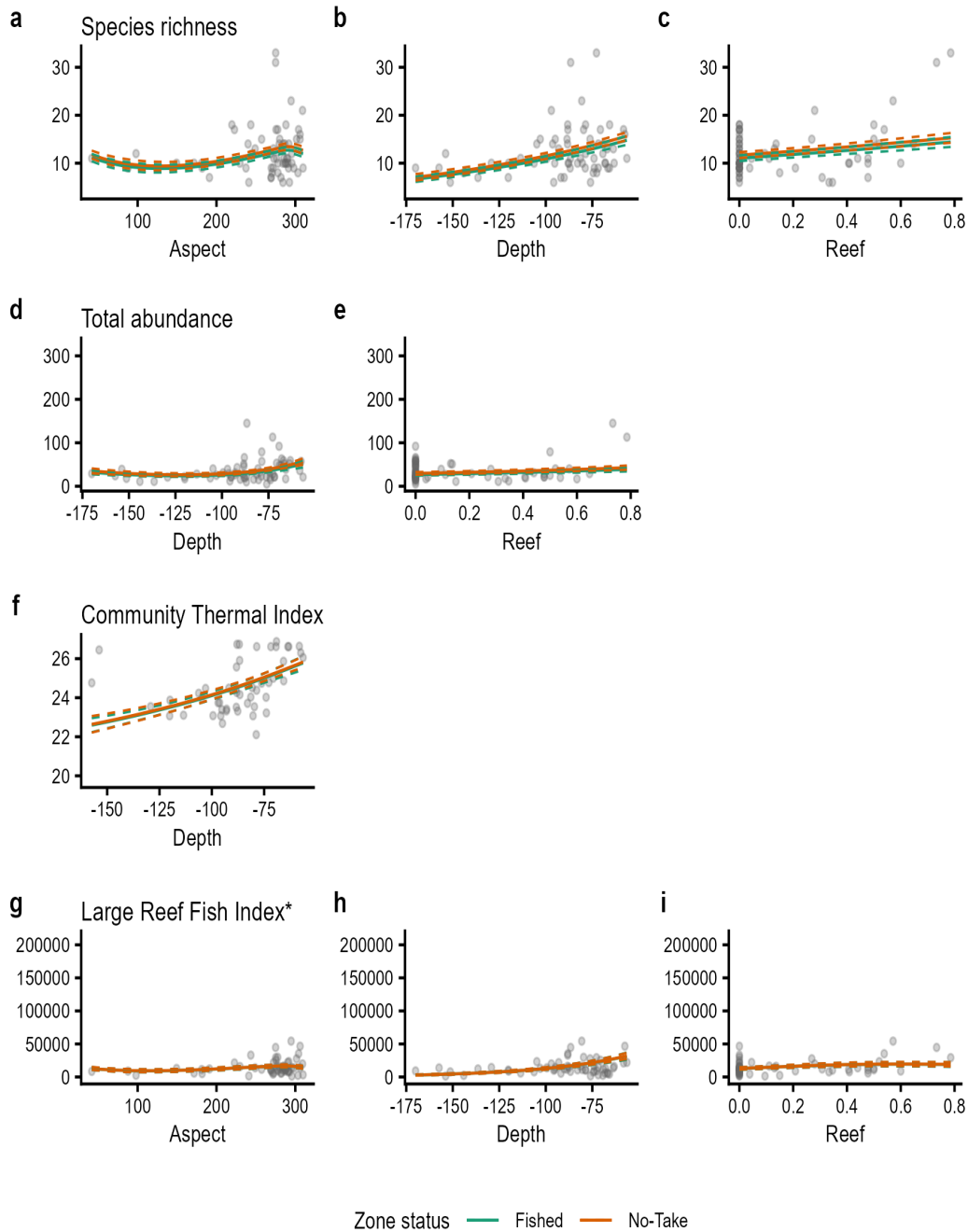


Figure C2.4.1: Plots of the most parsimonious model found to predict key demersal fish metrics in 2019. Individual panes display the predicted relationships between model covariates and each fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value and survey year held at 2019; line colour represents zone status (No-Take vs Fished). Dashed lines represent upper and lower standard error bounds around each prediction, and points represent the observed data for that year.

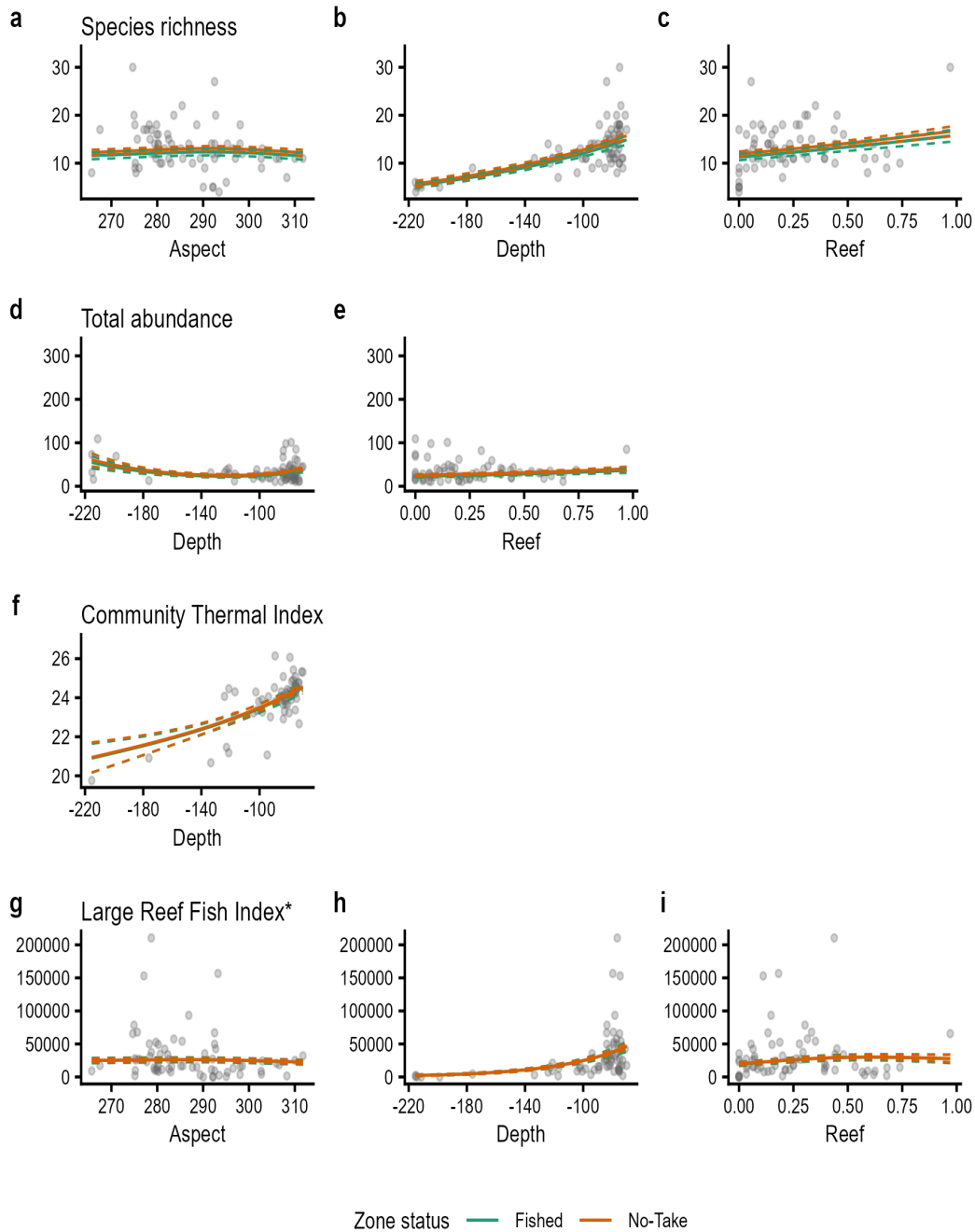


Figure C2.4.2: Plots of the most parsimonious model found to predict key demersal fish metrics in 2021. Individual panes display the predicted relationships between model covariates and each fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value and survey year held at 2021; line colour represents zone status (No-Take vs Fished). Dashed lines represent upper and lower standard error bounds around each prediction, and points represent the observed data for that year.

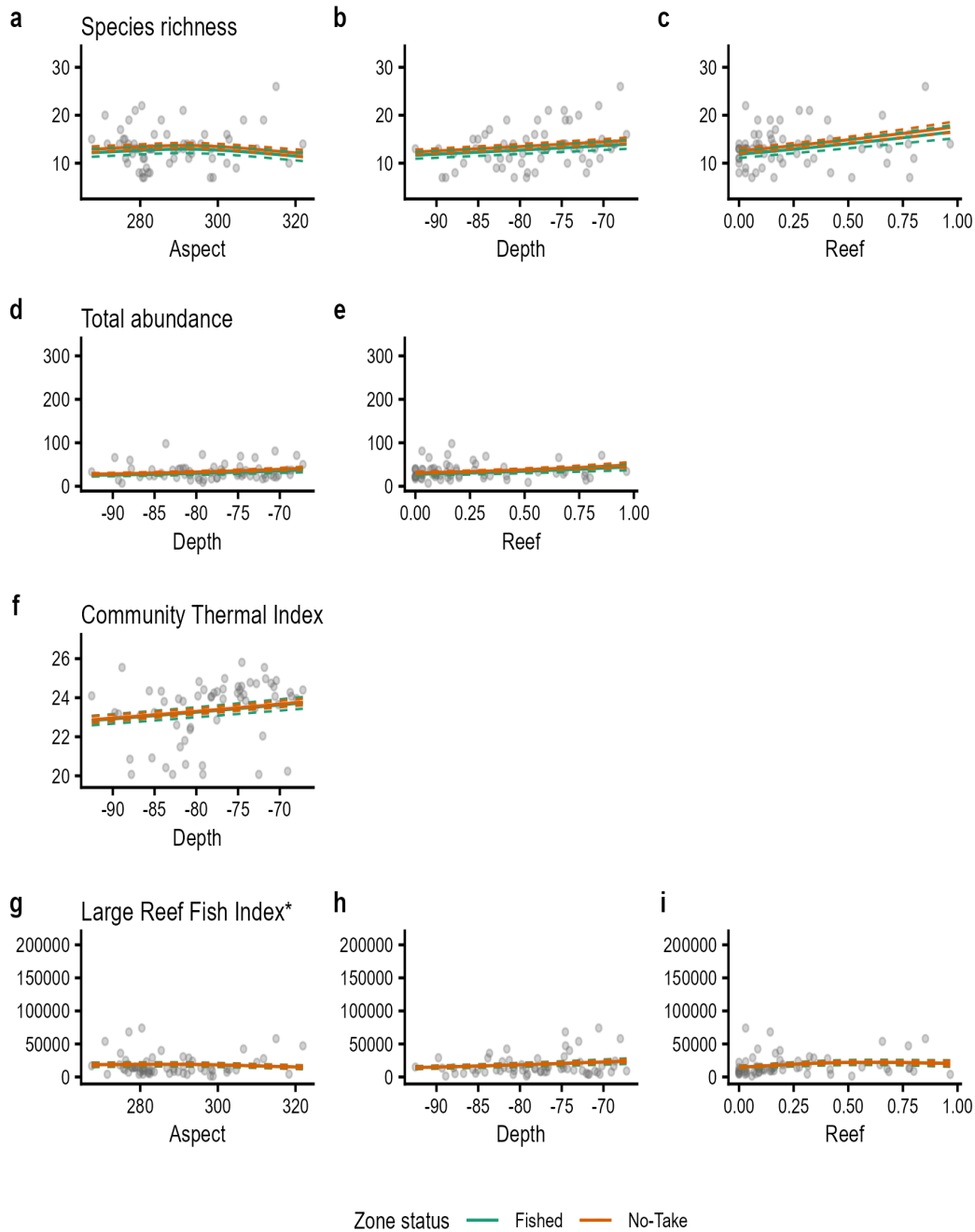


Figure C2.4.3: Plots of the most parsimonious model found to predict key demersal fish metrics in 2022. Individual panes display the predicted relationships between model covariates and each fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value and survey year held at 2022; line colour represents zone status (No-Take vs Fished). Dashed lines represent upper and lower standard error bounds around each prediction, and points represent the observed data for that year.

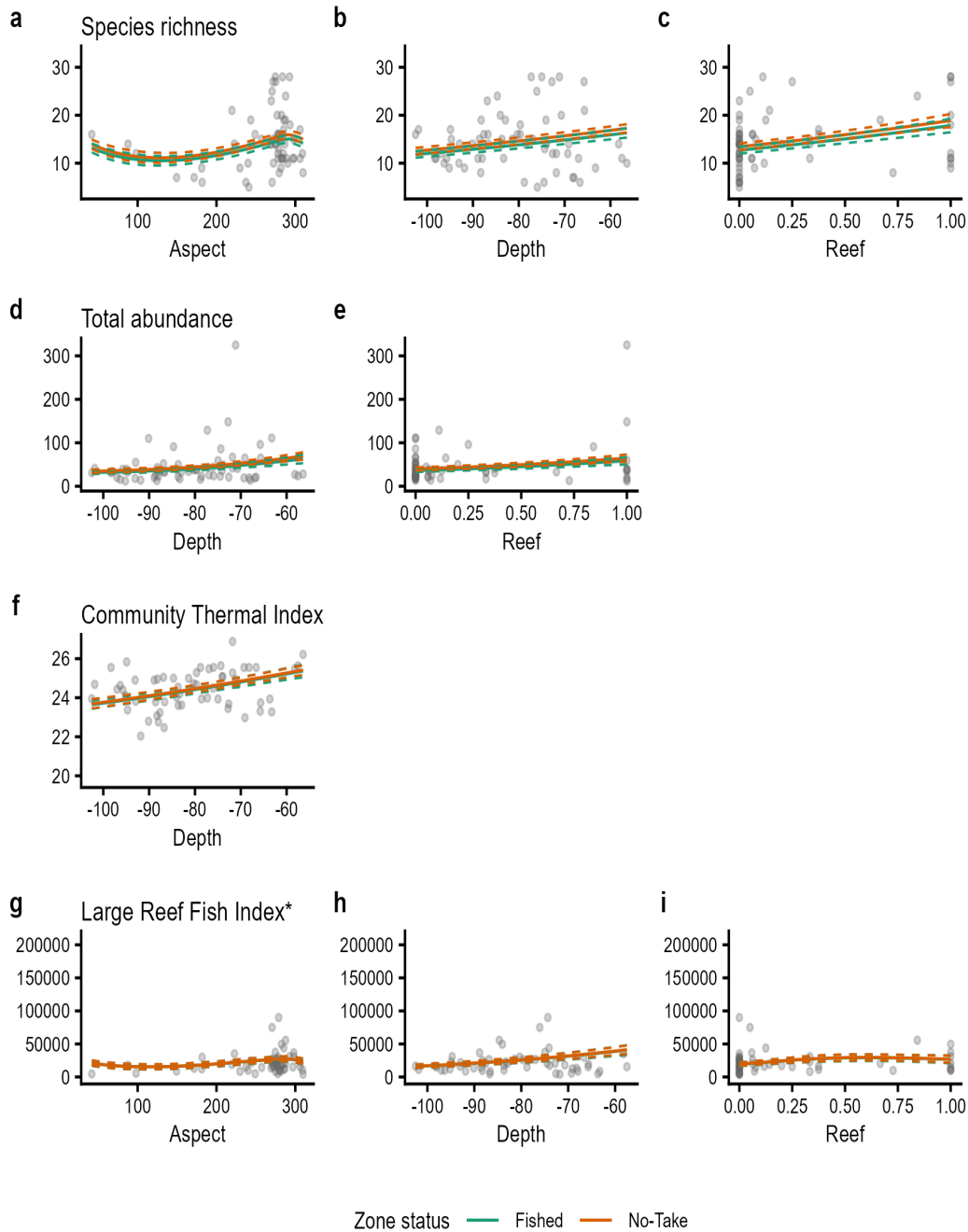


Figure C2.4.4: Plots of the most parsimonious model found to predict key demersal fish metrics in 2024. Individual panes display the predicted relationships between model covariates and each fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value and survey year held at 2024; line colour represents zone status (No-Take vs Fished). Dashed lines represent upper and lower standard error bounds around each prediction, and points represent the observed data for that year.